Skills Summary

Putting Rational and Irrational Numbers in Order

Use this reference while completing your worksheet or when studying.

Rational means writable as a fraction of whole numbers — every terminating or repeating decimal, and every root of a perfect square ($\sqrt{9} = 3$). **Irrational** means the decimal never terminates and never repeats: $\sqrt{2}$, $\sqrt{30}$, π.

One rule for the whole topic: you cannot order values written in different notations. Convert everything to a decimal first, carrying enough places that no two of them still look equal.

On the negative side the order reverses: $-2.3 < -2.25$, because further from zero is smaller.

1. Estimate a square root

Trap the radicand between the two perfect squares on either side of it; the root lies between their roots. Then refine by squaring candidate decimals — if the square is too big, come down. Squares worth knowing: $4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144$.

Example: Estimate $\sqrt{40}$ to the nearest hundredth.
Step 1. Trapping between perfect squares turns an unknown root into a range you can already read, so start there.
Step 2. $36 < 40 < 49$, so $6 < \sqrt{40} < 7$. Testing: $6.3^2 = 39.69$ (low) and $6.33^2 = 40.07$ (high), so the value is just under 6.33.
Rounded to the nearest hundredth, $\sqrt{40} = \mathbf{6.32}$.

Try it: Estimate $\sqrt{60}$ to the nearest hundredth.

check: **7.75**

2. Compare two quantities

Two ways, both quick. **Decimals:** convert both and read. **Squares:** for two positive values, squaring keeps the order, so compare $(\sqrt{7})^2 = 7$ against $2.7^2 = 7.29$ and skip the estimating entirely.

Example: Write $>$ or $<$ between $\sqrt{7}$ and 2.7.
Step 1. Both values are positive, so squaring is safe and it removes the radical without any decimal estimating at all.
Step 2. $(\sqrt{7})^2 = 7$ and $2.7^2 = 7.29$. Since 7 is less than 7.29, the same order holds before squaring.
So $\sqrt{7} < 2.7$.

Try it: Write $>$ or $<$ between $\sqrt{11}$ and 3.3.

check: $<$

3. Order a mixed list

Convert every value to a decimal, order the decimals, then write the answer back in the *original* symbols. Carry enough decimal places: if two values still look equal, you have not carried enough. Sort negatives and positives separately — every negative is less than every positive.

Example: Order $\sqrt{19}$, 4.3, $\dfrac{13}{3}$ from least to greatest.

Step 1. Three notations cannot be compared as they stand, so put all three into decimals before deciding anything.

Step 2. $\dfrac{13}{3} = 4.333$; $\sqrt{19} \approx 4.359$ (since $4.36^2 = 19.01$); 4.3 is given. The decimals rank 4.3, 4.333, 4.359.

Written in the original symbols: $\mathbf{4.3 < \frac{13}{3} < \sqrt{19}}$.

Try it: Order $\sqrt{8}$, 2.9, $\dfrac{14}{5}$ from least to greatest.

check: $\mathbf{\frac{14}{5} < \sqrt{8} < 2.9}$

(!) Watch out: π is not $\dfrac{22}{7}$ — they separate at the third decimal place, so an equals sign is always wrong there. And rounding too early is the main way these problems go wrong: to two places $\sqrt{2}$ and 1.41 look identical, and only a third place separates them.